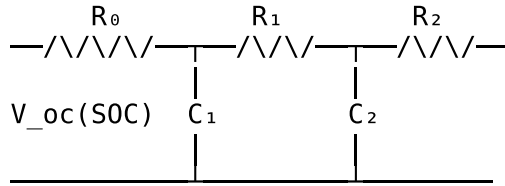


Mathematical Formulation of Exact Prediction Method

Battery Model Formulation

Equivalent Circuit Model:



Equations:

$$V_t = V_{oc}(SOC) - I \cdot R_0 - V_{p1} - V_{p2}$$

$$dV_{p1}/dt = I/C_1 - V_{p1}/(R_1 C_1)$$

$$dV_{p2}/dt = I/C_2 - V_{p2}/(R_2 C_2)$$

$$dSOC/dt = -\eta \cdot I / (3600 \cdot C_n \cdot SOH)$$

Power Constraint Formulation

Power Constraint Equation:

$$P = \eta_{conv} \cdot V_t \cdot I \\ = \eta_{conv} \cdot [V_{oc}(SOC) - I \cdot R_0 - V_{p1} - V_{p2}] \cdot I$$

This is a quadratic equation in I:

$$\eta_{conv} \cdot R_0 \cdot I^2 - \eta_{conv} \cdot [V_{oc} - V_{p1} - V_{p2}] \cdot I + P = 0$$

Solution:

$$I = [\eta_{conv} \cdot (V_{oc} - V_{p1} - V_{p2}) \pm \sqrt{(\eta_{conv}^2 \cdot (V_{oc} - V_{p1} - V_{p2})^2 - 4\eta_{conv} \cdot R_0 \cdot P)}] / (2\eta_{conv} \cdot R_0)$$

Continuous Integration Method

Continuous Integration Formulation:

Discharge time is obtained by integrating:

$$t_{end} = \int_{SOC_{min}}^{SOC_0} dt/dSOC \cdot dSOC$$

where:

$$dt/dSOC = - (3600 \cdot C_n \cdot SOH) / (\eta \cdot I(SOC))$$

and $I(SOC)$ is obtained by solving the power constraint equation at each SOC.

This is equivalent to solving the DAE system:

$$dy/dt = f(y, I), \quad g(y, I) = 0$$

$$\text{where } y = [SOC, V_{p1}, V_{p2}]^T$$

Numerical Solution Methods

Numerical Methods for Solution:

1. DAE Solver (Most Accurate):
 - Solves differential and algebraic equations simultaneously
 - Uses adaptive step size control
 - Error tolerance: $1e-8$ to $1e-10$
2. Shooting Method:
 - Converts to boundary value problem
 - Uses root finding to satisfy terminal conditions
3. Collocation Method:
 - Discretizes continuous problem
 - Solves large nonlinear system
4. Simple Integration (Approximate):
 - Assumes constant parameters
 - Fast but less accurate